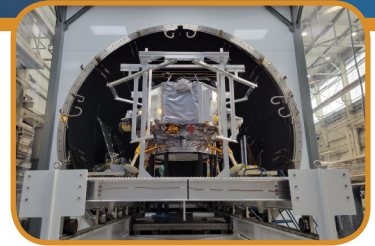




Marshall Space Flight Center's Environmental Test Facility Thermal Vacuum Test Chamber V-20

Engineering Solutions for Space Science and Exploration



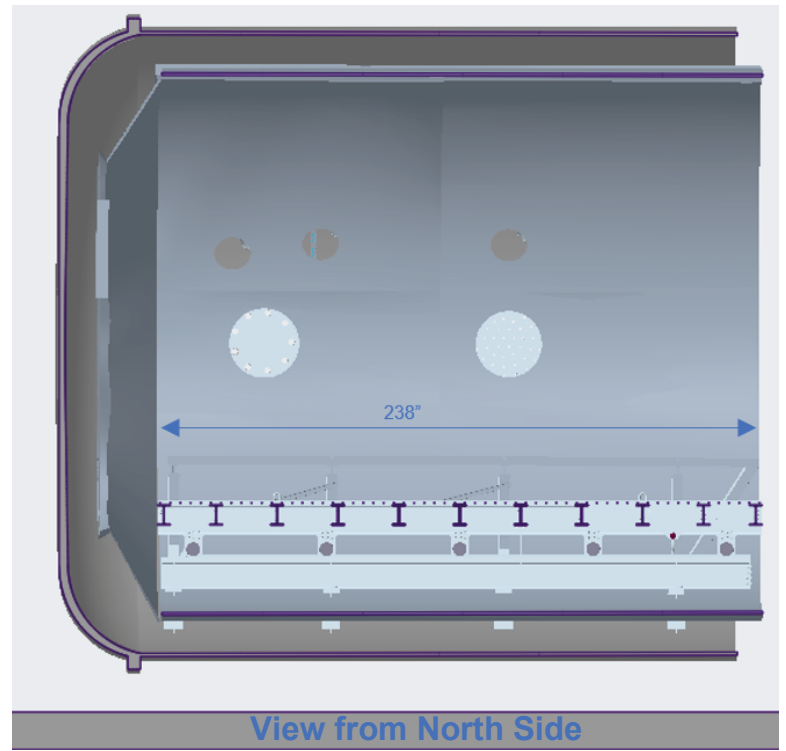
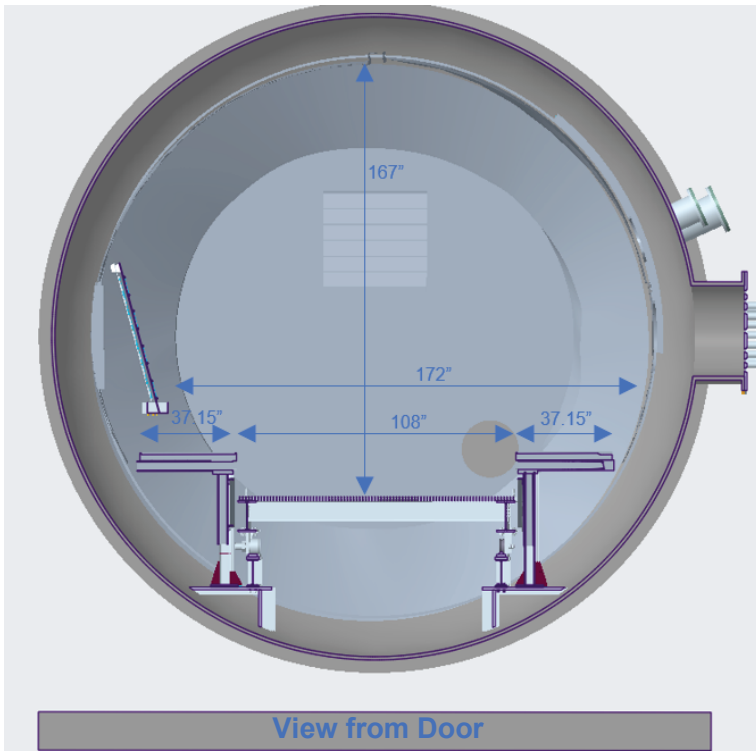
V-20 is a 20-foot Thermal Vacuum Chamber and Lunar Surface Simulator located at NASA Marshall Space Flight Center's Environmental Test Facility in Huntsville, Alabama. The capability of the thermal vacuum chamber was expanded in 2022 to simulate lunar & planetary environments in support of NASA's goal of returning to the Moon and NASA's Moon to Mars Program.

V-20 operates using two newly added 32-inch high-capacity diffusion pumps to provide the pumping capacity needed during lunar regolith operations such as sintering, additive manufacturing, and dust mitigation for mechanisms and structures. For flight tests or additional non-regolith pumping capacity, a 36" cryogenic pump along with three maglev turbomolecular pumps can additionally be used. The chamber is capable of achieving a vacuum level of 5×10^{-7} Torr.

V-20 has a thermal shroud installed, which can be filled with liquid Nitrogen (LN_2), or heated gaseous Nitrogen (GN_2). The chamber is equipped with 40-amp 0-240Vac power inputs with automatic control, with 12Vdc, 24Vdc, 208Vac, 240Vac, and 480Vac available. The chamber is also equipped with several blank flanges that penetrate the shroud to allow for test article feedthroughs for instrumentation & cabling. A chamber CAD file is available for more detailed information.

To aid with contamination concerns, two large 20 ft² cold plates have been added to the chamber, bolstering its ability to handle large gas loads. The chamber can also be equipped with a residual gas analyzer (RGA), Temperature Controlled Quartz Crystal Microbalance (TQCM), IR and color camera/video cameras, and gas purges.

The V-20 chamber is equipped with removable rails and loading cart rated at 60,000lbs, which allows for highly customizable regolith simulant beds or test articles. The cart is rolled into the chamber using a large winch located outside the chamber. Highly customizable fixturing, flanges, special test equipment, and feedthroughs are available per request. The chamber working area is enclosed inside a negative pressure dust containment tent.



Facility Capabilities:

Overall Dimension:	20' dia. x 28' deep
Data System:	National Instruments PXIE-1084 Chassis – 1hz baselined, 10hz max
Temperature Range:	-170C to +200C
Pressure:	5×10^{-7} torr
Thermocouples:	486 baselined
Thermal:	LN ₂ filled shroud, Heated shroud + IR Lamps
RGA:	Yes
TQCM:	Yes
Internal Camera:	IR and Color
CAD File:	Available upon request

Facility Applications

- Planetary Environment Simulator
- Large Lunar Regolith Simulant Manipulation
- Military, Civilian, and NASA Programs



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